Numerical Methods: Root Finding and Interpolation

Self-explanatory teaching notes (from zero background)

1. What Does Finding a Root Mean?

A **root** (or zero) of a function $f(x)$ is a value α such that

$$f(\alpha) = 0.$$

Geometrically, it is where the graph cuts the x-axis. Numerical methods do not guess blindly; they generate a **sequence** of approximations that converges to the root.

2. Newton–Raphson Method

Idea

Near a point, a smooth curve behaves like its tangent. We replace the curve with its tangent line and use the x-intercept.

Derivation

Tangent at $(x_n, f(x_n))$:

$$y - f(x_n) = f'(x_n)(x - x_n)$$

Set $y = 0$:

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

Worked Example

Solve $x^2 - 2 = 0$, $f'(x) = 2x$, take $x_0 = 1.5$.

Iteration Table

n	x_n	$f(x_n)$
0	1.5000	0.2500
1	1.4167	0.0069
2	1.4142	0.0000

Pros and Cons

Pros: Quadratic convergence (very fast).
Cons: Requires derivative; poor guess may diverge.

3. Secant Method

Idea

Approximate the derivative using two previous points.

Formula

$$x_{n+1} = x_n - f(x_n)\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

Worked Example

$f(x) = x^2 - 2$, $x_0 = 1$, $x_1 = 2$.

Iteration Table

n	x_n	$f(x_n)$
0	1.0000	-1.0000
1	2.0000	2.0000
2	1.3333	-0.2222
3	1.4000	-0.0400

Pros and Cons

Pros: No derivative needed.
Cons: No guaranteed convergence.

4. False Position (Regula Falsi)

Idea

Keep the root bracketed but use a secant instead of midpoint.

Formula

$$x = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

Worked Example

$f(x) = x^2 - 2$, $a = 1$, $b = 2$.

Iteration Table

n	a	b	x
1	1.0000	2.0000	1.3333
2	1.3333	2.0000	1.4000

Pros and Cons

Pros: Guaranteed convergence.
Cons: May stagnate.

5. Muller's Method

Idea

Fit a quadratic through three points and take its root.

Formula

$$x_{n+1} = x_n - \frac{2c}{b \pm \sqrt{b^2 - 4ac}}$$

Worked Example

$f(x) = x^2 - 2$, $x_0 = 1$, $x_1 = 1.5$, $x_2 = 2$ gives $x \approx 1.414$.

Pros and Cons

Pros: Finds complex roots.
Cons: Algebraically heavy.

6. Horner's Method

Purpose

Efficient polynomial evaluation and basis of root algorithms.

Worked Example

Evaluate $P(x) = 2x^3 - 6x^2 + 2x - 1$ at $x = 3$.

Horner Table

$$\begin{array}{c|cccc} 3 & 2 & -6 & 2 & -1 \\ & & 6 & 0 & 6 \\ \hline & 2 & 0 & 2 & 5 \end{array}$$

Thus $P(3) = 5$.

7. Cubic Spline Interpolation

Idea

Use piecewise cubic polynomials with smooth first and second derivatives.

Spline Form

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

Conditions

- Interpolation at data points
- Continuity of first derivative
- Continuity of second derivative
- Natural spline: $S''(x_0) = S''(x_n) = 0$

Tridiagonal System

For equal spacing h:

$$hm_{i-1} + 4hm_i + hm_{i+1} = 6\left(\frac{y_{i+1} - y_i}{h} - \frac{y_i - y_{i-1}}{h}\right)$$

How to Use (4 lines)

1. Form equations for second derivatives.
2. Solve tridiagonal system.
3. Construct each cubic polynomial.
4. Evaluate spline at desired x.

Example Question

Given $(0, 1), (1, 2), (2, 0)$, construct the natural cubic spline and estimate $f(1.5)$.

8. Final Wisdom

Newton is fastest, Secant is practical, False Position is safe, Muller is flexible, Horner is foundational, and Cubic Splines preserve smoothness without oscillation.